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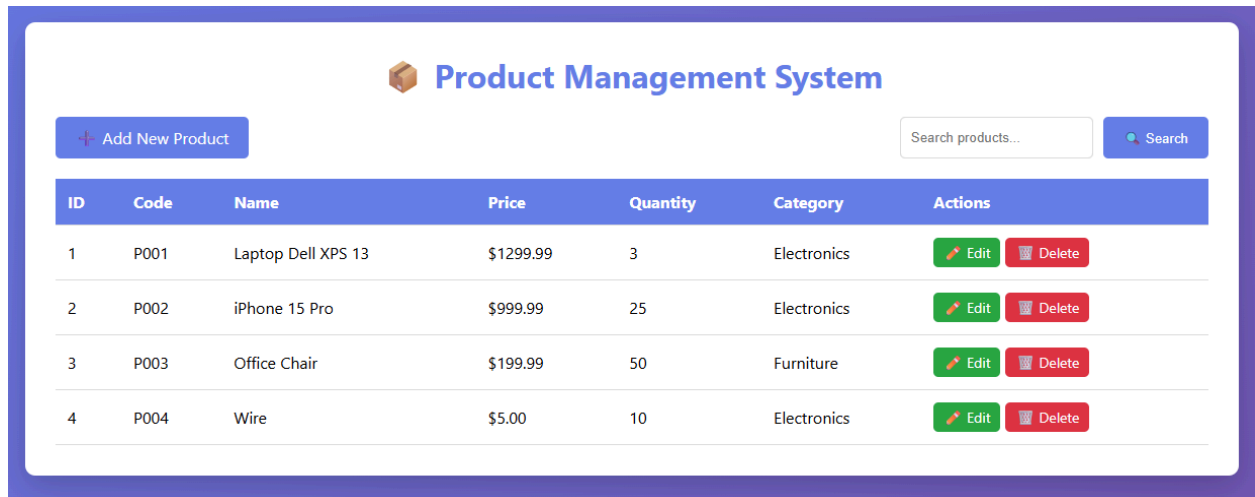
Lab 7 Practice

GitHub link:

<https://github.com/ntthanhbang/WebDevClass/tree/main/product-management/src/main>

List all products:

<http://localhost:8080/products>



ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	3	Electronics	<button>Edit</button> <button>Delete</button>
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	<button>Edit</button> <button>Delete</button>
3	P003	Office Chair	\$199.99	50	Furniture	<button>Edit</button> <button>Delete</button>
4	P004	Wire	\$5.00	10	Electronics	<button>Edit</button> <button>Delete</button>

Explain:

- Browser requests /products
- ProductController.listProducts() is executed
 @GetMapping
 public String listProducts(Model model)
- Controller calls service:
 productService.getAllProducts()
- Service calls repository:
 productRepository.findAll()
- JPA generates SQL:
 SELECT * FROM products
- Repository returns List<Product>
- Controller adds list to Model:
 model.addAttribute("products", products)
- Controller returns view name: "product-list"
- Thymeleaf searches for:

src/main/resources/templates/product-list.html

- <http://localhost:8080/products/new>

After adding:

Product Management System

Product updated successfully!

 Add New Product

Search products...

 Search

ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	3	Electronics	 Edit  Delete
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	 Edit  Delete
3	P003	Office Chair	\$199.99	50	Furniture	 Edit  Delete
4	P004	Wire	\$5.00	10	Electronics	 Edit  Delete
5	P005	Earphones	\$12.00	4	Electronics	 Edit  Delete

Explain:

- Browser requests /products/new
ProductController.showNewForm() runs
@GetMapping("/new")
- Controller creates a blank Product
Product product = new Product();
model.addAttribute("product", product);
- Returns view: "product-form"
- Thymeleaf loads: product-form.html
- HTML displays a form (input fields)

Edit product (ID=1):

<http://localhost:8080/products/edit/1>



Edit Product

Product Code *

Product Name *

Price (\$) *

Quantity *

Category *

Electronics

Description

High-performance laptop

 Save Product

 Cancel

After editing:



Product Management System

Product updated successfully!

 Add New Product

 Search

ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	7	Electronics	 Edit  Delete
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	 Edit  Delete
3	P003	Office Chair	\$199.99	50	Furniture	 Edit  Delete
4	P004	Wire	\$5.00	10	Electronics	 Edit  Delete
5	P005	Earphones	\$12.00	4	Electronics	 Edit  Delete

Explain:

- Browser requests /products/edit/1
@PathVariable Long id extracts id = 1
- Controller runs:
 productService.getProductById(1)
- Service calls repository:
 productRepository.findById(1)
- JPA generates SQL:
 SELECT * FROM products WHERE id = 1
- Repository returns Optional<Product>
- If product exists: Add product to Model. Return product-form
- If product does NOT exist: Set redirect message. Redirect to /products

Search products:

<http://localhost:8080/products/search?keyword=laptop>


Product Management System

 Add New Product

 Search

ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	7	Electronics	 Edit  Delete


Product Management System

 Add New Product

 Search

No products found. Add your first product!

Explain:

- Browser requests /products/search?keyword=laptop
@RequestParam("keyword") extracts "laptop"
- Controller calls service:
 productService.searchProducts("laptop")
- Service calls repository:
 productRepository.findByNameContaining("laptop")
- Spring Data JPA automatically generates SQL:
 SELECT * FROM products
 WHERE name LIKE '%laptop%'

- Repository returns List<Product>
- Controller adds it to Model:
 model.addAttribute("products", products);
- Return view "product-list"

Delete product (ID=5):

<http://localhost:8080/products/delete/5>



Product Management System

Product deleted successfully!

[Add New Product](#)

[Search](#)

ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	7	Electronics	Edit Delete
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	Edit Delete
3	P003	Office Chair	\$199.99	50	Furniture	Edit Delete
4	P004	Wire	\$5.00	10	Electronics	Edit Delete

Explain:

- Browser requests /products/delete/5
- Controller extracts id = 5
- Controller calls service:
 productService.deleteProduct(5)
- Service calls repository:
 productRepository.deleteById(5)
- JPA generates SQL:
 DELETE FROM products WHERE id = 5
- Controller adds success message (flash attribute)
- Redirect back to: "redirect:/products"